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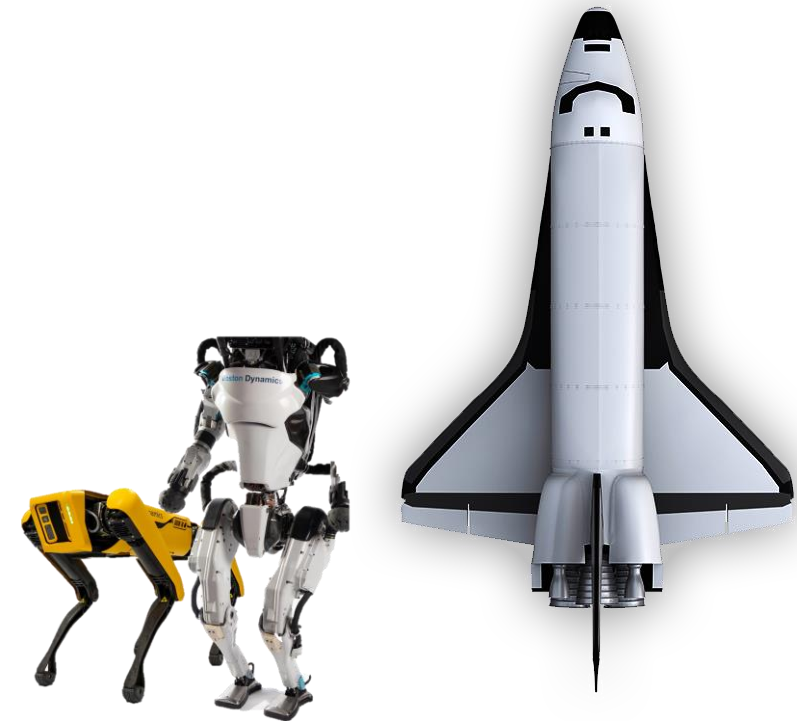
ADVANCED CONTROL SYSTEM DESIGN

Lecture 4

State-Space Analysis



University of
Nottingham
UK | CHINA | MALAYSIA



Lecture Outline

01

Eigenvalues

02

Diagonalization

03

Diagonalization – Distinct Real Roots

04

Diagonalization – Repeated Real Roots

Eigenvalues

The eigenvalues (characteristic roots) of an $n \times n$ matrix $\mathbf{A}$ are the roots of the **characteristic equation**:

$$|\lambda \mathbf{I} - \mathbf{A}| = 0$$

Similar to transfer functions, the roots of the system can be used to investigate the **stability**. For instance, for a second-order system with the following characteristic equation:

$$a\lambda^2 + b\lambda + c = 0$$

The eigenvalues can be determined using the quadratic formula:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The parameters of the quadratic formula can also be used to determine the **damping** behavior of a **stable system**.

Eigenvalues

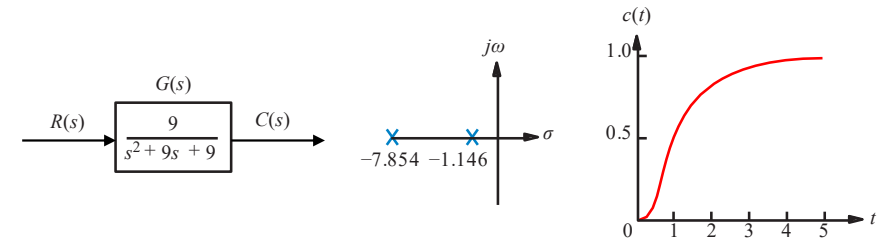
01

The system is **overdamped** when:

$$b^2 > 4ac$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -9 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 9 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



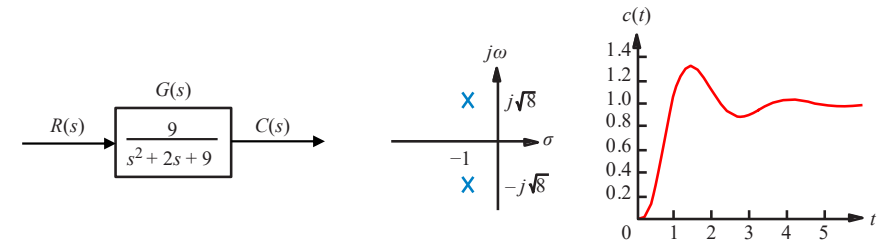
02

The system is **underdamped** when:

$$b^2 < 4ac$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -9 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 9 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



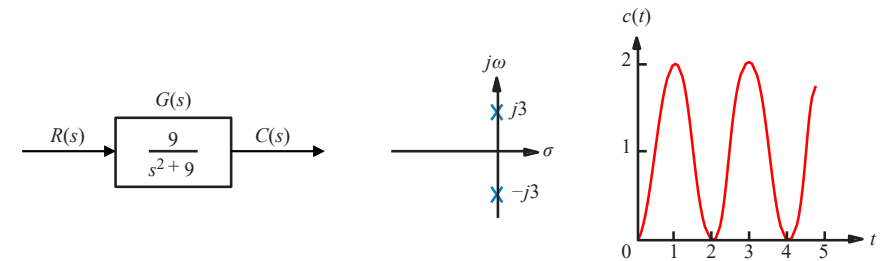
03

The system is **undamped** when:

$$b = 0$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -9 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 9 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



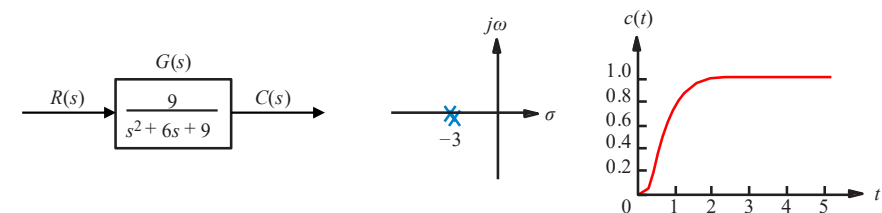
04

The system is **critically damped** when:

$$b^2 = 4ac$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -9 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 9 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



Eigenvalues

EXAMPLE 1

Find the eigenvalues of the following **A** matrix:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}$$

$$|\lambda \mathbf{I} - \mathbf{A}| = \begin{vmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{vmatrix} - \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{vmatrix} = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{vmatrix} = \lambda^3 + 6\lambda^2 + 11\lambda + 6$$

$$(\lambda + 1)(\lambda + 2)(\lambda + 3) = 0$$

The eigenvalues of **A** are the roots of the characteristic equation, or -1 , -2 , and -3 .

Eigenvalues

EXERCISE 1

Find the eigenvalues of the following $\mathbf{A}$ matrices and comment on the systems:

$$\mathbf{A}_1 = \begin{bmatrix} 0 & 0 & -120 \\ 1 & 0 & -74 \\ 0 & 1 & -15 \end{bmatrix}$$

$$\mathbf{A}_2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & 1 & -6 \end{bmatrix}$$

$$\mathbf{A}_3 = \begin{bmatrix} -7 & -12 & -10 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{A}_4 = \begin{bmatrix} 1 & 5 & 6 \\ 2 & 5 & 7 \\ -4 & -2 & -8 \end{bmatrix}$$

Diagonalization

WHAT IS IT?

A similarity transformation to convert the state-space representation into a form that conveniently shows the system eigenvalues.

ADVANTAGES

- Expose the system roots.
- Decouple the system parameter matrix without changing the input-output relationship.

POSSIBILITIES

For a system with an $n \times n$ $\mathbf{A}$ matrix, there are 3 possible scenarios: **distinct real roots**, **repeated real roots**, or **complex roots**. We will focus on the first 2 cases in this module.

Diagonalization – Distinct Real Roots

- If an $n \times n$ $\mathbf{A}$ matrix with **real distinct eigenvalues** is given by:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}$$

- We define the transformation $\mathbf{x} = \mathbf{P}\mathbf{z}$, where:

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 & \cdots & \lambda_n^2 \\ \vdots & \vdots & \vdots & & \vdots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \lambda_3^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix}$$

- $\lambda_1, \lambda_2, \lambda_3 = n$ **real distinct** eigenvalues of $\mathbf{A}$.
- We can use $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ to transform $\mathbf{A}$ into the diagonal matrix:

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{bmatrix} \lambda_1 & 0 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 & 0 \\ 0 & 0 & \lambda_3 & 0 & 0 \\ 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 0 & \lambda_n \end{bmatrix}$$

- Thus, we can represent the diagonalized system as follows:

$$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{z} + \mathbf{P}^{-1}\mathbf{B}u$$

$$y = \mathbf{C}\mathbf{P}\mathbf{z} + \mathbf{D}u$$

Diagonalization – Repeated Real Roots

- If an $n \times n$ $\mathbf{A}$ matrix with r real repeated eigenvalues is given by:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}$$

- We define the transformation $\mathbf{x} = \mathbf{S}\mathbf{z}$, where:

$$\mathbf{S} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 1 & \cdots & 1 \\ \lambda_1 & 1 & 0 & \cdots & \lambda_{r+1} & \cdots & \lambda_n \\ \lambda_1^2 & 2\lambda_1 & 1 & \cdots & \lambda_{r+1}^2 & \cdots & \lambda_n^2 \\ \lambda_1^3 & 3\lambda_1^2 & 3\lambda_1 & \cdots & \lambda_{r+1}^3 & \cdots & \lambda_n^3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \lambda_1^{n-1} & (n-1)\lambda_1^{n-2} & \frac{n(n-1)}{2}\lambda_1^{n-3} & \cdots & \lambda_{r+1}^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix}$$

- For simplicity, $\mathbf{S}$ is given as follows when $n = 3$:

$$\mathbf{S} = \begin{bmatrix} 1 & 0 & 1 \\ \lambda_1 & 1 & \lambda_3 \\ \lambda_1^2 & 2\lambda_1 & \lambda_3^2 \end{bmatrix}$$

- It is not possible to perform complete diagonalization in this case, but we can perform block diagonalization, as follows:

$$\mathbf{S}^{-1}\mathbf{A}\mathbf{S} = \begin{bmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \quad \leftarrow \text{Jordan}$$

$$\dot{\mathbf{z}} = \mathbf{S}^{-1}\mathbf{A}\mathbf{S}\mathbf{z} + \mathbf{S}^{-1}\mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{S}\mathbf{z} + \mathbf{D}\mathbf{u}$$

Notes on Diagonalization

01

The diagonalization approaches presented earlier are based on Sylvester's interpolation formula and Vandermonde's matrix.

02

Matrices $\mathbf{P}$ and $\mathbf{S}$ can be found using other approaches, such as finding the eigenvectors associated with the eigenvalues using standard mathematical approaches.

03

The eigenvectors approach can also be used to diagonalize systems with complex eigenvalues. However, this will increase the complexity of the solution due to the presence of complex number. In addition, having complex numbers in the state-space is not preferred in many cases. Thus, the state-space needs to be represented using real numbers using several approaches, such as Euler's formula.

Diagonalization

EXAMPLE 2

Perform diagonal transformation on the state-space equations below:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Diagonalization

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x}$$

$$|\lambda \mathbf{I} - \mathbf{A}| = \begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{vmatrix} = \lambda^3 + 6\lambda^2 + 11\lambda + 6$$

$$\lambda_1 = -1, \lambda_2 = -2, \lambda_3 = -3$$

Since the eigenvalues are real and distinct, then:

$$\mathbf{x} = \mathbf{P}\mathbf{z}$$

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

$$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{z} + \mathbf{P}^{-1}\mathbf{B}u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} 3 & 2.5 & 0.5 \\ -3 & -4 & -1 \\ 1 & 1.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 3 & 2.5 & 0.5 \\ -3 & -4 & -1 \\ 1 & 1.5 & 0.5 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} u$$

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} u$$

$$y = \mathbf{C}\mathbf{P}\mathbf{z}$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Diagonalization

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 6 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$



$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Diagonalization

EXERCISE 2

Perform diagonal transformation on the state-space equations below:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -75 & -55 & -13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} u$$

$$y = \begin{bmatrix} 4 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Controllability in State-Space

Next Week ...

**DO YOU
HAVE ANY
QUESTIONS**